Task 1: Review of Programming Basics

Python and C# Console Applications

Programming Exercises – Complete Solutions

November 2026

Overview

This report presents working console-based solutions for three exercises in both Python and C#. Each solution reads input, computes a result, and prints it to the console. Input validation prevents invalid operations and grades.

1 Simple Two-Number Calculator

Goal: Read two numbers and an operator (+, -, *, /), then display the result. Division by zero is handled explicitly.

Python

```python
try:
    first = float(input("First number: "))
    second = float(input("Second number: "))
    operator = input("Operation (+, -, *, /): ").strip()

    if operator == "+":
        result = first + second
    elif operator == "-":
        result = first - second
    elif operator == "*":
        result = first * second
    elif operator == "/":
        if second == 0:
            raise ZeroDivisionError("Cannot divide by zero.")
        result = first / second
    else:
        raise ValueError("Unknown operation.")

    print(f"Result: {result:g}")
except (ValueError, ZeroDivisionError) as error:
    print(f"Error: {error}")
```

C#

```csharp
using System;
using System.Globalization;

Console.Write("First number: ");
bool firstOk = double.TryParse(Console.ReadLine(),
    NumberStyles.Float, CultureInfo.InvariantCulture, out double a);
```

```
Console.Write("Second number: ");
bool secondOk = double.TryParse(Console.ReadLine(),
    NumberStyles.Float, CultureInfo.InvariantCulture, out double b);
Console.Write("Operation (+, -, *, /): ");
string op = Console.ReadLine()?.Trim() ?? "";

if (!firstOk || !secondOk)
{
    Console.WriteLine("Error: Invalid number.");
}
else if (op == "/" && b == 0)
{
    Console.WriteLine("Error: Cannot divide by zero.");
}
else
{
    double result;
    if (op == "+") result = a + b;
    else if (op == "-") result = a - b;
    else if (op == "*") result = a * b;
    else if (op == "/") result = a / b;
    else
    {
        Console.WriteLine("Error: Unknown operation.");
        return;
    }
    Console.WriteLine($"Result: {result:G}");
}
```

Example: Inputs: `10`, `20`, `+`. Output: `Result: 30`.

2 Temperature Converter

Goal: Select C to convert Celsius to Fahrenheit, or F to convert Fahrenheit to Celsius. The formulas are

$$F = 1.8C + 32, \qquad C = \frac{F - 32}{1.8}.$$

Python

```
try:
    direction = input("Convert from C or F? ").strip().upper()
    if direction not in ("C", "F"):
        raise ValueError("Choose C or F.")
    temperature = float(input("Temperature: "))

    if direction == "C":
        converted = temperature * 1.8 + 32
        print(f"{temperature:g} C = {converted:g} F")
    else:
        converted = (temperature - 32) / 1.8
        print(f"{temperature:g} F = {converted:g} C")
except ValueError as error:
    print(f"Error: {error}")
```

C#

```
using System;
using System.Globalization;

Console.Write("Convert from C or F? ");
string direction = (Console.ReadLine() ?? "").Trim().ToUpper();
if (direction != "C" && direction != "F")
{
    Console.WriteLine("Error: Choose C or F.");
    return;
}
Console.Write("Temperature: ");
if (!double.TryParse(Console.ReadLine(), NumberStyles.Float,
    CultureInfo.InvariantCulture, out double temperature))
{
    Console.WriteLine("Error: Invalid temperature.");
    return;
}
if (direction == "C")
{
    double fahrenheit = temperature * 1.8 + 32;
    Console.WriteLine($"{temperature:G} C = {fahrenheit:G} F");
}
else
{
    double celsius = (temperature - 32) / 1.8;
    Console.WriteLine($"{temperature:G} F = {celsius:G} C");
}
```

Example: Inputs: `C`, `0`. Output: `0 C = 32 F`.

3 Student Grade Point Average

Goal: Read a positive number of grades on the 1–6 scale, calculate their arithmetic mean, and print whether the student passed (average at least 3.0).

Python

```
try:
    count = int(input("Number of grades: "))
    if count <= 0:
        raise ValueError("Enter at least one grade.")

    total = 0.0
    for index in range(count):
        grade = float(input(f"Grade {index + 1} (1-6): "))
        if not 1 <= grade <= 6:
            raise ValueError("Grades must be between 1 and 6.")
        total += grade

    average = total / count
    print(f"Average: {average:.2f}")
    if average >= 3.0:
        print("The student passed.")
    else:
        print("The student did not pass.")
except ValueError as error:
    print(f"Error: {error}")
```

C#

```
using System;
using System.Globalization;

Console.Write("Number of grades: ");
if (!int.TryParse(Console.ReadLine(), out int count) || count <= 0)
{
    Console.WriteLine("Error: Enter a positive number of grades.");
    return;
}
double total = 0;
for (int i = 0; i < count; i++)
{
    Console.Write($"Grade {i + 1} (1-6): ");
    if (!double.TryParse(Console.ReadLine(), NumberStyles.Float,
        CultureInfo.InvariantCulture, out double grade) ||
        grade < 1 || grade > 6)
    {
        Console.WriteLine("Error: Grades must be between 1 and 6.");
        return;
    }
    total += grade;
}
double average = total / count;
Console.WriteLine($"Average: {average:F2}");
if (average >= 3.0)
    Console.WriteLine("The student passed.");
else
    Console.WriteLine("The student did not pass.");
```

Example: Inputs: `4`, `5`, `3`, `4`, `2`. Output: `Average: 3.50` followed by `The student passed.`

Execution and Submission

Each code sample is a standalone console application. Save Python examples as `task1.py`, `task2.py`, and `task3.py`, and run with `python task1.py` (similarly for the other tasks). For C#, create a separate console project for each task using `dotnet new console`, replace its `Program.cs` with the corresponding code, and execute `dotnet run`. C# examples use top-level statements (C# 9 or later). Numeric input in C# uses a dot as decimal separator.

For submission, push the source files to GitHub, preferably with separate Python and C# directories and a README containing run instructions. Use a feature branch (for example, `feature/programming-basic`) and merge it into `main` after testing. A GitHub repository link must be created and submitted separately; this document does not represent a published repository.